

Voice Cloning Models

Evaluation and Comparison

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Vision

To Generate multilingual audio books in the author's own voice in minutes.

Data Preparation

- Sample voices stored in
data/Sample_voice<number>/
- Multiple files per speaker to enhance cloning quality
- Text pages for audiobook generation

Python Library Analysis

- Speechbrain for speaker embeddings
- TTS (Coqui / VITS / XTTS) for open-source TTS
- ElevenLabs and OpenAI TTS APIs for industry solutions
- Librosa, Torchaudio, Pandas, Matplotlib, Seaborn for analysis

Open-source TTS Model Research

- Models tried: your_tts, XTTS-v2, VITS
- Technical struggles: Multi-file input handling, embedding similarity calculation, PyTorch version issues
- Faced more technical issues with XTTS-v2. So excluded it

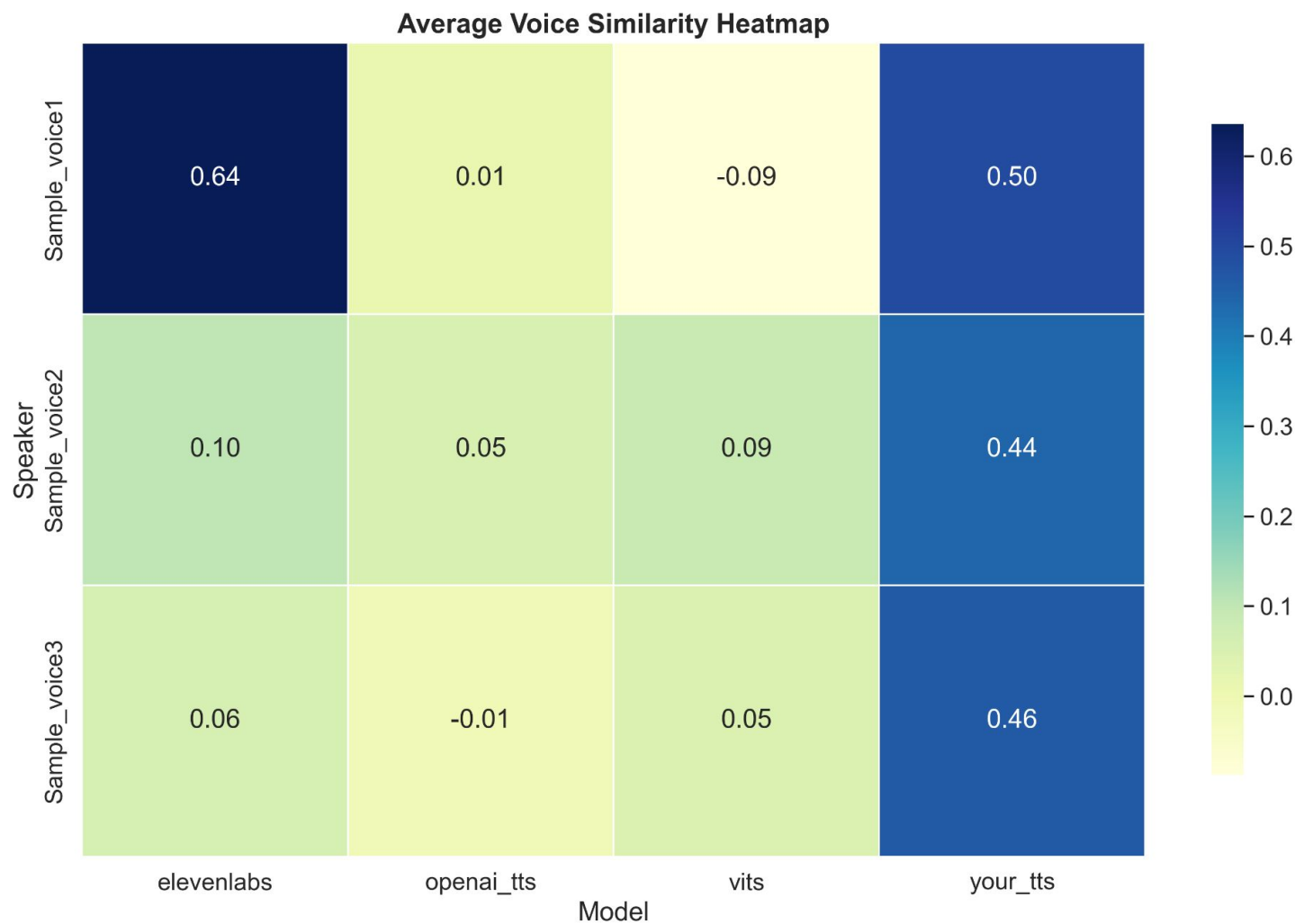
Industry Models & Challenges

- Models: ElevenLabs, OpenAI TTS
- Challenges: API limitations, cost per minute, multi-voice cloning constraints(Especially with Eleven labs)

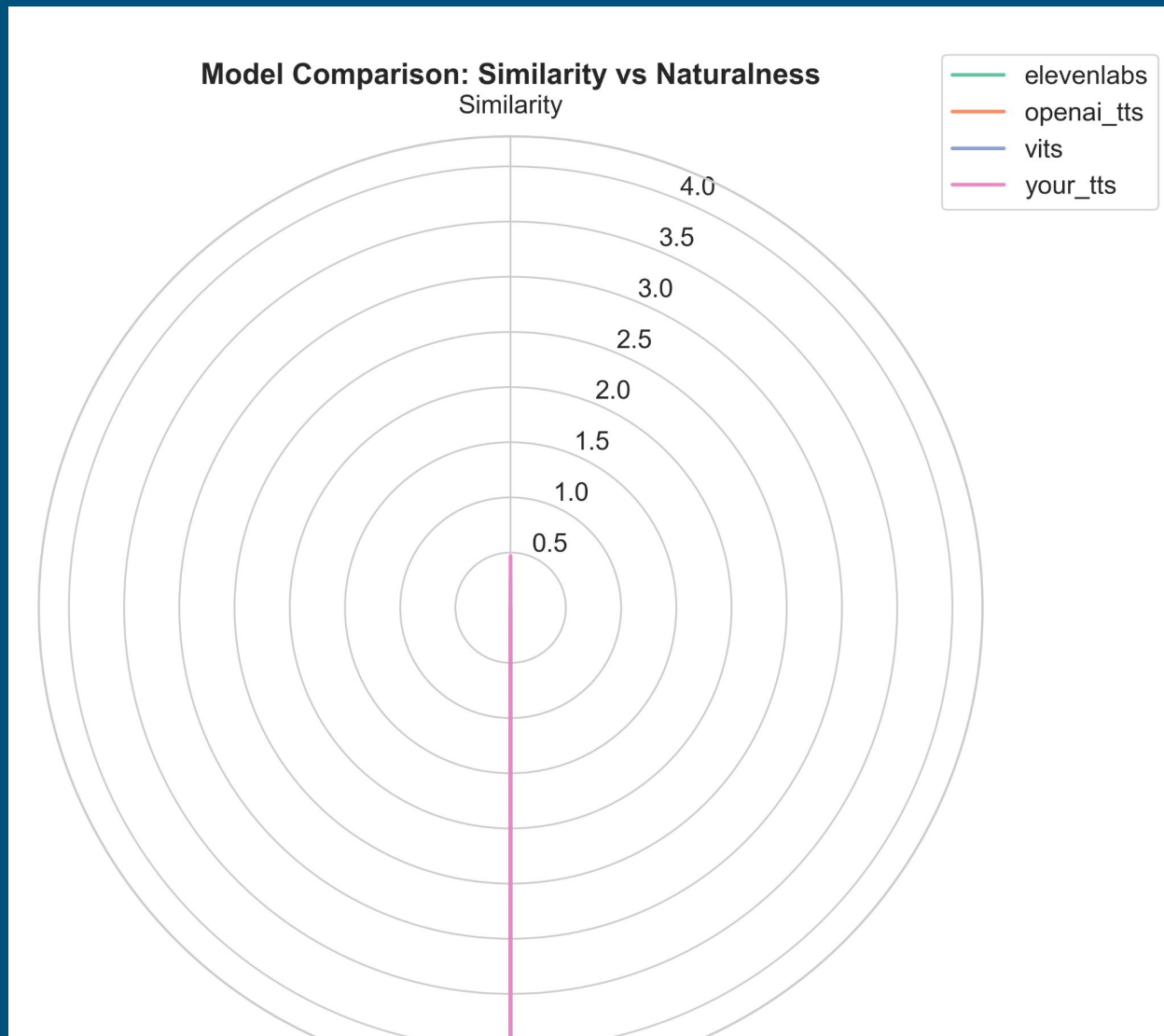
Automation for Evaluation

- Metrics: Voice cloning clarity (Cosine Similarity) and Naturalness (MOS / acoustic metrics)
- Benchmarking code for single vs multiple voice inputs
- Visualizations for similarity, naturalness, inference time, and cost

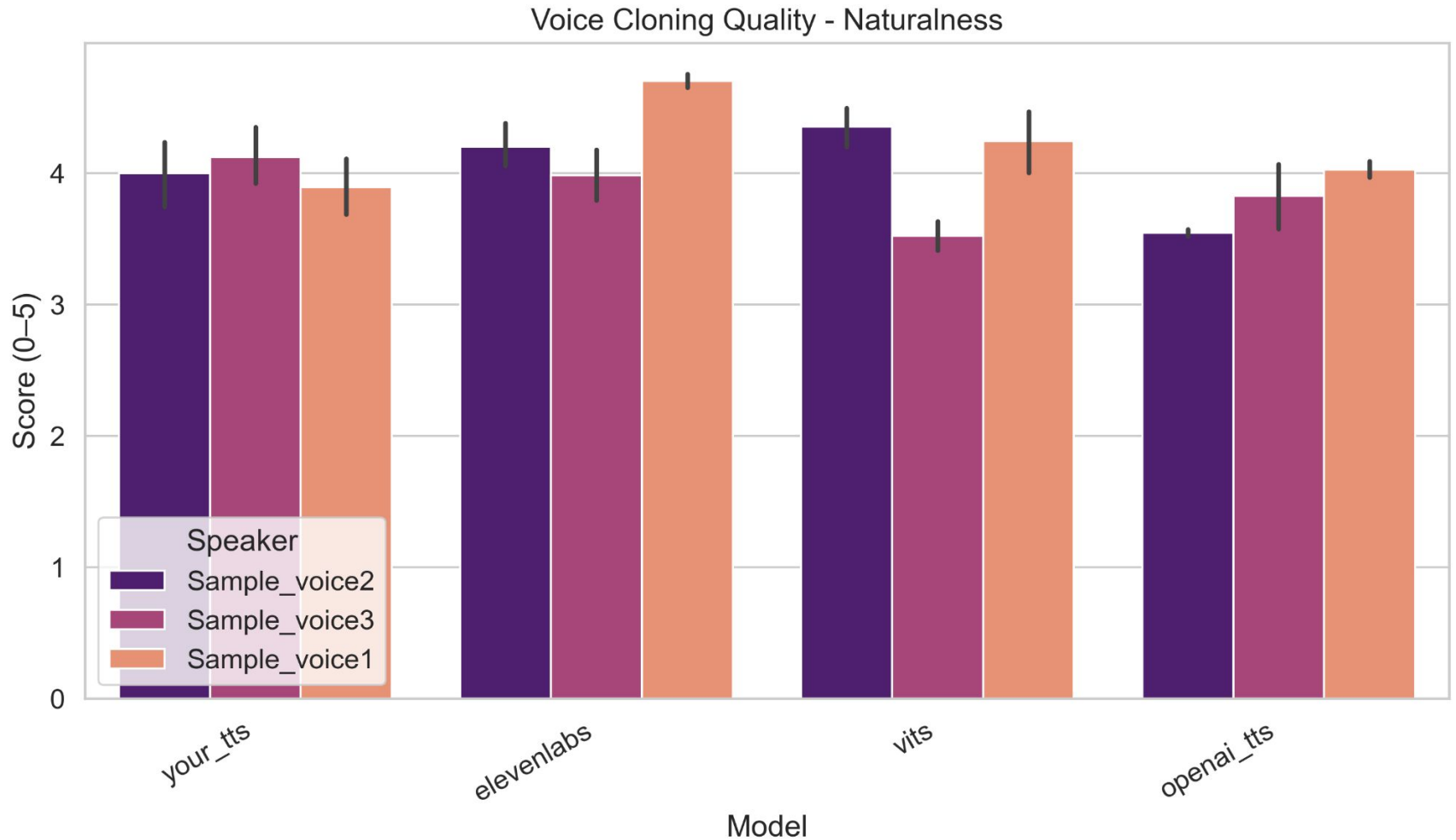
Average Voice similarity



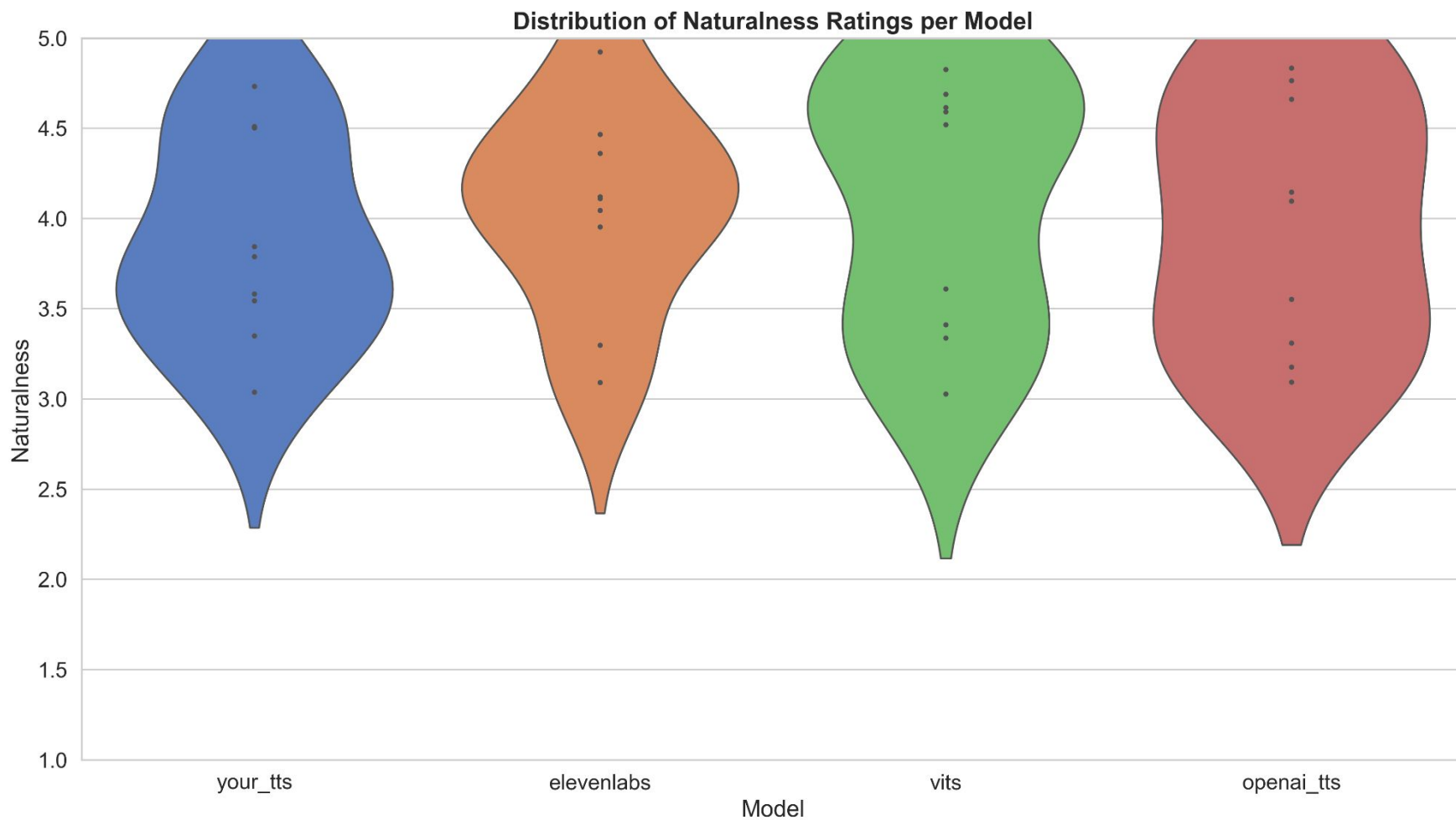
Similarity vs Naturalness



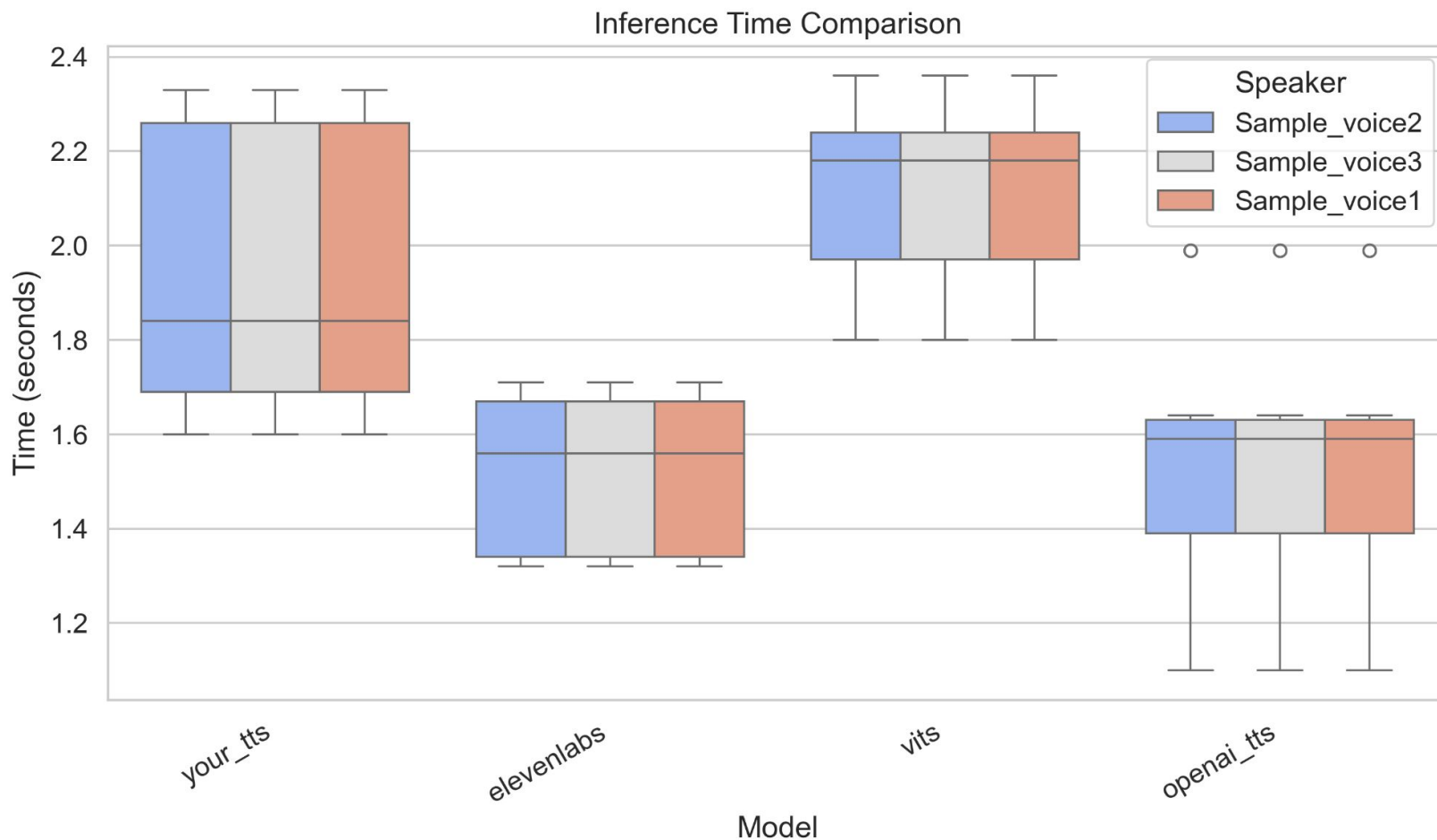
Voice cloning quality



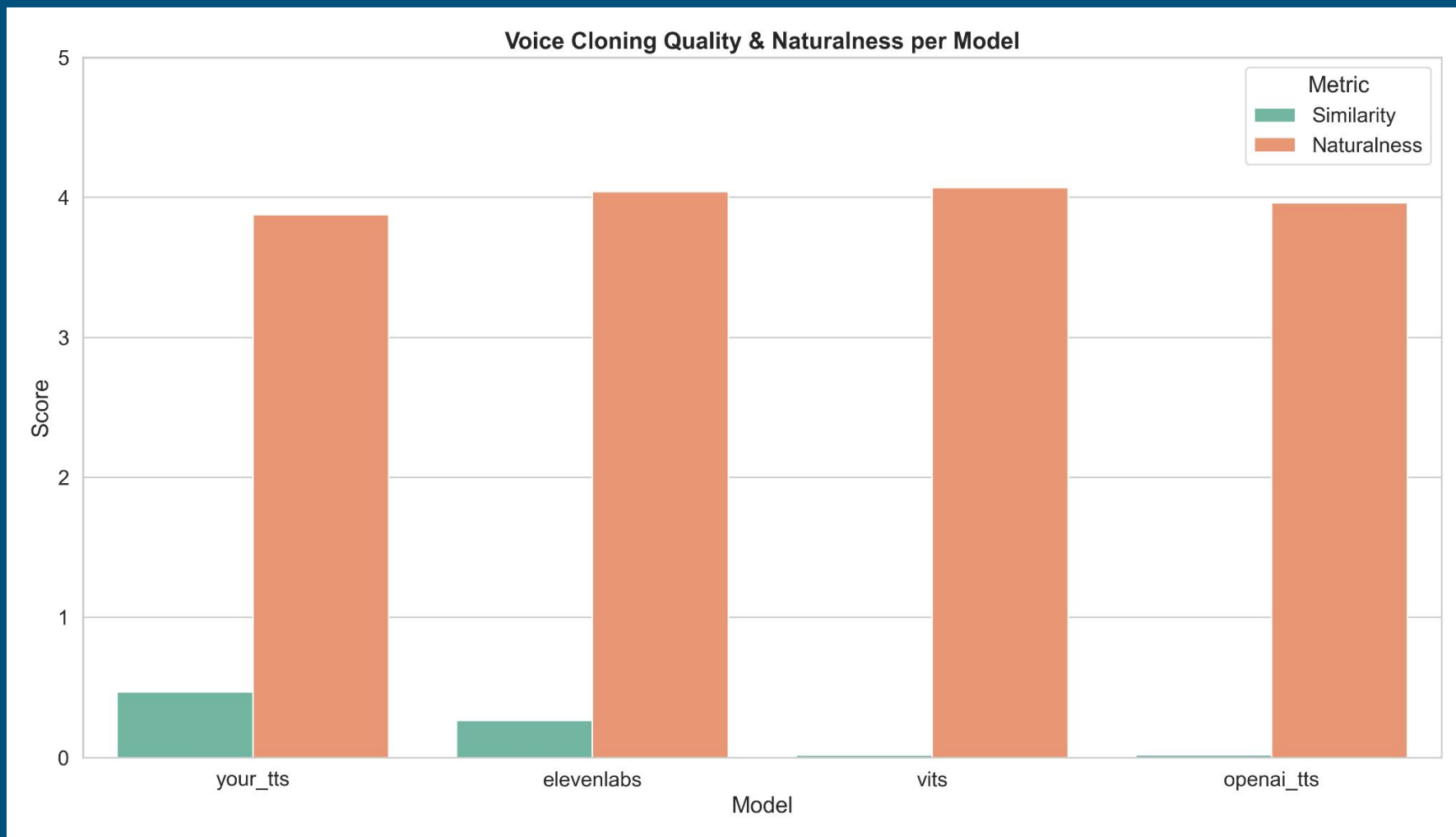
Distribution of Naturalness



Inference Time Comparison

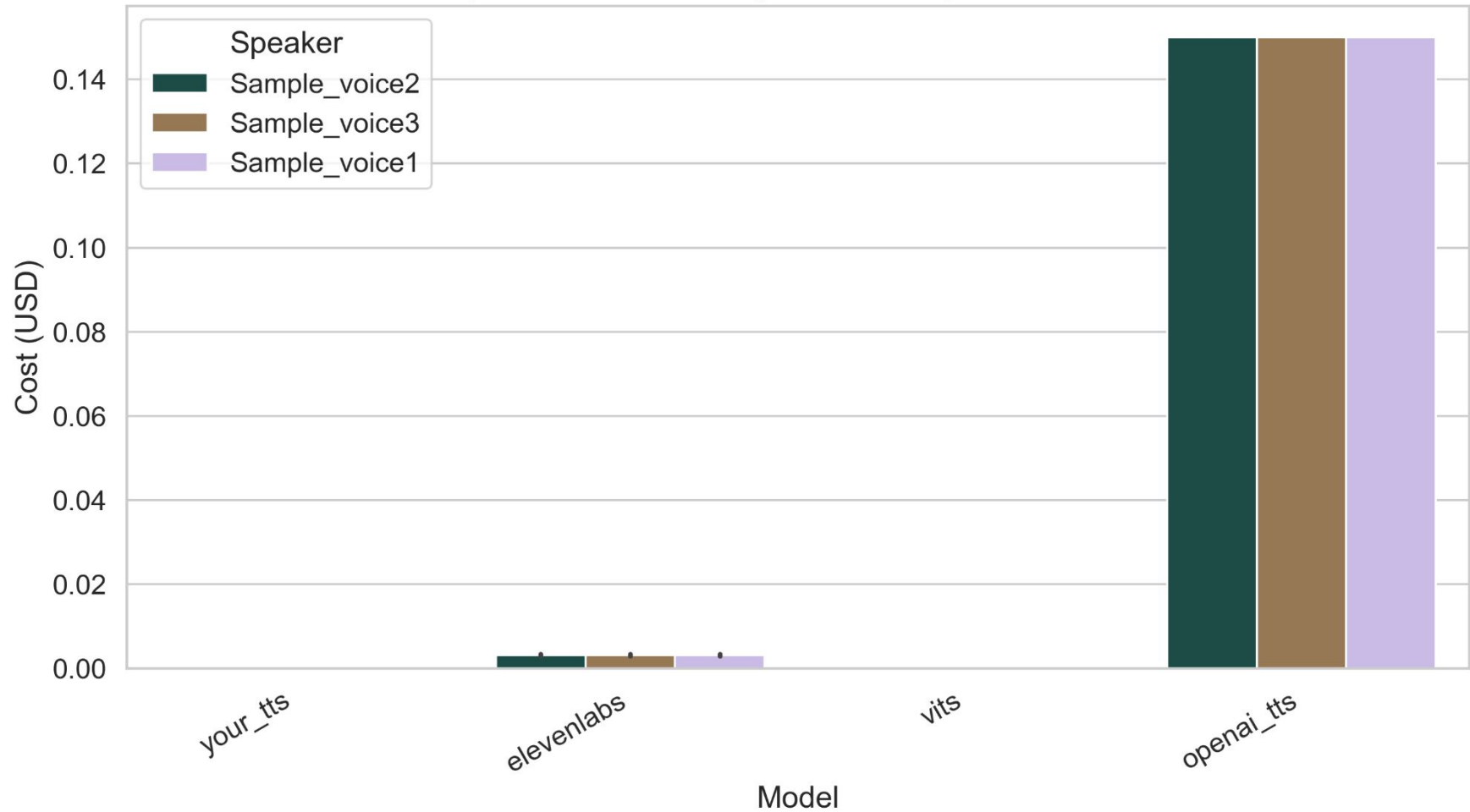


Voice cloning Quality and Naturalness



Cost per 1000 characters

Cost per 1000 Characters (No cost for Open source models)



Conclusion

- ElevenLabs is highly efficient but costly for multiple voice cloning
- Open-source models offer flexibility but require more technical setup
- Vision: Optimize voice cloning pipeline for audiobook production

Vision for Audio books

- Develop open-source multilingual models
- Enable authors to **publish multilingual audiobooks in minutes**
- Personalize narration using **author's own voice**
- Reduce production cost and time for **high-quality audio content**
- Enhance accessibility: **readers in multiple languages**, visually impaired audiences
- Reduce cost and make technology accessible

How This Capstone Helps

- Benchmarks **open-source & industry TTS / voice cloning models**
- Supports **multi-sample speaker inputs** for accurate voice cloning
- Generates **page-wise audio outputs**, ready for audiobook workflows
- Measures **cost, speed, and voice similarity** for informed model selection
- Provides a foundation for **scalable, automated audiobook generation**

Thank you Note

- Begin understanding **how voice models work**
- Identify **areas to focus on** for improvement
- Learn how to **fine-tune models and resolve challenges**

This is just the **beginning of my journey** in exploring voice cloning and audiobook generation.